

INTERNSHIP PROJECT REPORT

KINEMATIC AND DYNAMIC ANALYSIS OF A CUSTOM 4-DOF ROBOTIC MANIPULATOR

Submitted to the Bharat Space Education Research Centre (BSERC)
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DECLARATION

We hereby declare that the project report entitled "Kinematic and Dynamic Analysis of a Custom 4-DOF Robotic Manipulator" submitted to the Bharat Space Education Research Centre (BSERC) for the Def-Space Summer Internship program is a record of original work done by us. The simulation models, trajectory analyses, and structural geometries presented in this report have not been submitted elsewhere for any other internship or academic credit.

Dated: August 30, 2026

A handwritten signature in blue ink that reads "G. Ananth Srivasthava". The signature is written in a cursive, flowing style.

(G Ananth Srivasthava)

ACKNOWLEDGEMENT

We express our deepest gratitude to the Bharat Space Education Research Centre (BSERC) for providing the platform and resources necessary to conduct this study.

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ABSTRACT

Rapid prototyping of robotic manipulators for defense and space ecosystems requires rigorous validation of both spatial kinematics and contact dynamics prior to physical fabrication. This project details the computational modeling and dynamic analysis of a custom 4-Degree-of-Freedom (4-DOF) robotic manipulator using MATLAB and Simscape Multibody. The system incorporates a mathematically defined U-shaped end effector, modeled as an extruded solid, to simulate future payload manipulation. The manipulator was evaluated across three distinct operational phases: (1) mapping the 3D operational workspace using continuous harmonic sine wave actuation, (2) validating rigid body environmental interactions using a spatial contact force block and a convex hull geometry to simulate table impact, and (3) generating smooth, mathematically continuous point-to-point step trajectories by applying second-order polynomial filtering to mitigate infinite torque singularities. The results successfully validated the forward kinematics against standard Denavit-Hartenberg parameters, demonstrated accurate dynamic buckling under unactuated collision, and produced S-curve kinematics critical for stable space robotics deployment.

Supplementary Video Data: The simulated animations for all three test cases can be viewed via the following Google Drive repository:

[drive.google.com/drive/folders/1zJAEvfGcTCBtwOHBINdMYBtr0rXGBJW0?usp=drive_link].

Contents

Declaration	ii
Acknowledgement	iii
Abstract	iv
Table of Contents	v
List of Figures	vi
List of Tables	vi
List of Symbols and Abbreviations	vii
1 INTRODUCTION	1
1.1 Background	1
1.2 Project Objectives	1
2 METHODOLOGY	2
2.1 Mechanism Configuration and Structural Links	2
2.2 Custom End Effector Geometry	2
2.3 Case 1: Forward Kinematics Setup (Sine Wave Input)	3
2.4 Case 2: Contact Dynamics Setup (Collision)	3
2.5 Case 3: S-Curve Trajectory Generation (Step Input)	4
3 RESULTS AND DISCUSSION	5
3.1 Forward Kinematics Workspace (Sine Wave)	5
3.2 Dynamic Environmental Collision	6
3.3 Smooth Trajectory Generation and Motion Tracking	6
3.4 Kinematic Validation (D-H Parameters)	7
4 CONCLUSION AND FUTURE SCOPE	8
5 APPENDICES	9

List of Figures

2.1	Constructed Simulink Architecture for Sine Wave Forward Kinematics	3
2.2	Modified Simulink Architecture for Environmental Collision Physics	4
2.3	Constructed Architecture for S-Curve Filtering and Trajectory Generation . . .	4
3.1	Simulink Scope Output Showing Harmonic Response Signals Over Time . . .	5
3.2	Obtained 3D Workspace Trajectory from Harmonic Inputs	6
3.3	Obtained Trajectory Exhibiting Physical Buckling During Table Impact	6
3.4	Obtained S-Curve Spatial Trajectory and Motion Tracking Scope Output from Filtered Step Inputs	7

List of Tables

3.1	D-H Parameters of the 4-DOF Manipulator	7
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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Description
DOF	Degree of Freedom
D-H	Denavit-Hartenberg
BSERC	Bharat Space Education Research Centre
IK	Inverse Kinematics
a_i	Link Length
α_i	Link Twist Angle
d_i	Joint Offset
θ_i	Joint Angle (q)
τ	Time Constant
f_n	Normal Force

INTRODUCTION

1.1 Background

The deployment of robotic manipulators in aerospace and defense environments demands absolute precision and reliability. Mathematical errors in trajectory planning or physical collisions during autonomous operations can lead to catastrophic hardware failure. Traditionally, robotic design relies on physical prototyping, which is both time-consuming and expensive. Simscape Multibody provides a robust physical modeling environment to mathematically validate kinematics, torques, and contact forces before physical materials are allocated.

1.2 Project Objectives

A custom lightweight, cylindrical 4-DOF robotic arm was developed to analyze motion planning, map 3D workspace limits, and resolve mathematical solver singularities caused by infinite torque demands. The primary objectives of this simulation study are:

- To design a functional 4-DOF kinematic chain with a custom mathematically defined U-shaped end effector suitable for payload extraction.
- To map the forward kinematics workspace using continuous harmonic sine wave actuation.
- To simulate physical environmental collision by manipulating rigid body joint actuation states (passive vs. active control).
- To generate smooth, continuous step trajectories by applying second-order polynomial filtering to raw input signals, preventing solver crashes.

Chapter 2

METHODOLOGY

Every individual parameter, solver setting, and geometrical dimension configured within the MATLAB/Simscape environment is detailed in this section to ensure full reproducibility of the robotic simulation.

2.1 Mechanism Configuration and Structural Links

The simulation environment was initialized using the standard Simscape physical network blocks: Solver Configuration, World Frame, and Mechanism Configuration. The global reference frame was established at $Z = 0$ to serve as the ground origin.

The manipulator's structural links were constructed utilizing parameterized Cylindrical Solid blocks with explicit standard SI dimensions:

- **Base Link:** Diameter of 0.10 m and length (L_{base}) of 0.10 m.
- **Second Link (Upper Arm):** Length (L_{upper}) of 0.20 m and diameter of 0.04 m.
- **Third Link (Lower Arm):** Length (L_{lower}) of 0.10 m and diameter of 0.03 m.

These individual masses were connected sequentially via four Revolute Joint blocks representing the Base, Shoulder, Elbow, and Wrist. To establish the orthogonal axes of rotation necessary for a functional 4-DOF spatial envelope, Rigid Transform blocks were inserted between each joint and link with precise rotational offsets ($+X$ at 90° and -90°). The translational Z -offsets correspond directly to the link offset ($d_1 = L_{base}$) parameter in standard Denavit-Hartenberg kinematic formulations.

2.2 Custom End Effector Geometry

A custom U-shaped gripper was modeled to serve as the manipulator's active tool point with a depth (extrusion length) of 0.04 m. Instead of importing external CAD step files, an Extruded Solid block was utilized for computational efficiency. The geometry was defined mathematically using a 2D cross-sectional matrix provided in a strict counter-clockwise order:

$$[-2, 0; 2, 0; 2, 5; 1, 5; 1, 1; -1, 1; -1, 5; -2, 5] \text{ cm} \quad (2.1)$$

The local reference frame (the Proximal port) was established on the bottom bounding face of the U-shape (the base "palm") with axes aligned to $+Y$ and $+X$ to project the tool's functional centerline forward from the wrist joint.

Because the U-shaped gripper is a custom concave polygon, the Simscape collision engine export settings were explicitly set to **Convex Hull**, creating an invisible physical boundary across the fingertips for standard G (Geometry) port collision detection.

2.3 Case 1: Forward Kinematics Setup (Sine Wave Input)

To trace the 3D workspace envelope, continuous harmonic inputs were utilized.

- **Actuation:** All four Revolute Joints were configured via the Actuation menu with Motion set to **Provided by Input** and Torque set to **Automatically Computed**.
- **Inputs:** Four Sine Wave blocks were generated using default default frequency parameters (1 rad/s) and an amplitude fixed at $\pi/2$ radians.
- **Sensing:** A Transform Sensor block connected between the World Frame and the End Effector routed X, Y , and Z coordinates to a To Workspace block to record the trajectory array.

Figure 2.1 illustrates the construction of the block diagram for this continuous kinematic sweep, linking the mathematical Sine Wave sources directly into the q actuation ports.

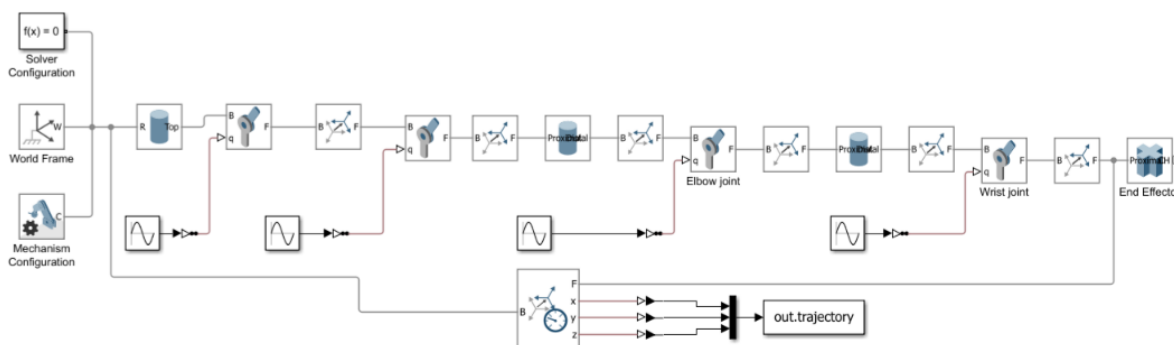


Figure 2.1: Constructed Simulink Architecture for Sine Wave Forward Kinematics

2.4 Case 2: Contact Dynamics Setup (Collision)

To simulate a structural crash and validate rigid body physics, a flat table was modeled using a Brick Solid block ($0.50 \times 0.50 \times 0.02$ m) elevated via a Rigid Transform to $Z = 0.30$ m. To prevent infinite torque violations, the outer Elbow and Wrist joints were modified with Motion set to **Automatically Computed** and Torque to **None** (passive mechanism). A Spatial Contact Force block bridged the End Effector's convex hull and the Table to measure the normal force (f_n) impact spike.

Chapter 3

RESULTS AND DISCUSSION

The data collected effectively validates the operational mechanics of the robot across all three simulation phases.

3.1 Forward Kinematics Workspace (Sine Wave)

The application of harmonic inputs resulted in a dense spatial cloud, confirming the mathematical compounding of coordinate frames.

Additionally, the temporal behavior and harmonic tracking performance of the system are illustrated by the Simulink Scope output in Figure 3.1. The curves display continuous, phase-shifted sinusoidal waveforms representing the individual joint or spatial coordinate variations across a 10-second simulation span. This confirms that the harmonic inputs maintain smooth, continuous derivative profiles without abrupt velocity or acceleration discontinuities.

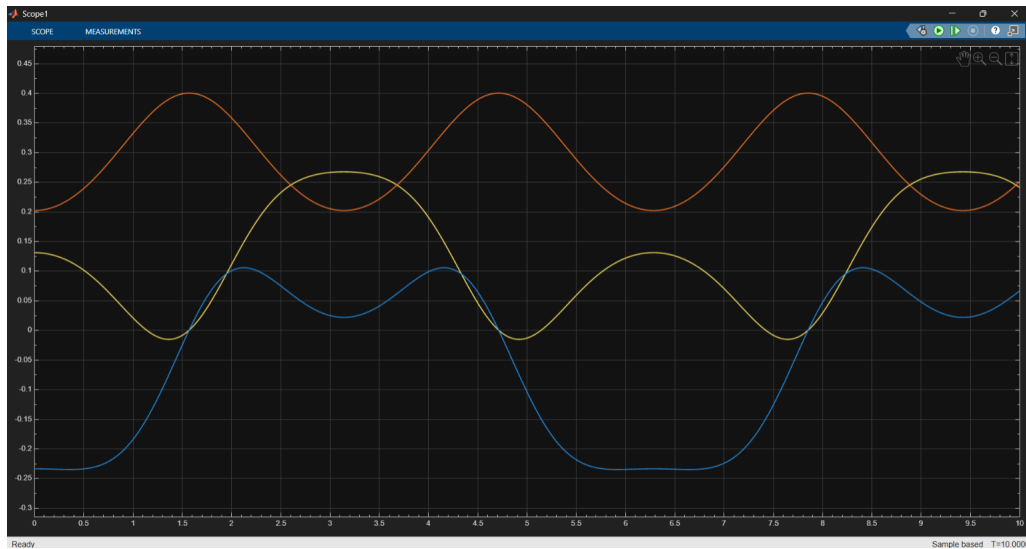


Figure 3.1: Simulink Scope Output Showing Harmonic Response Signals Over Time

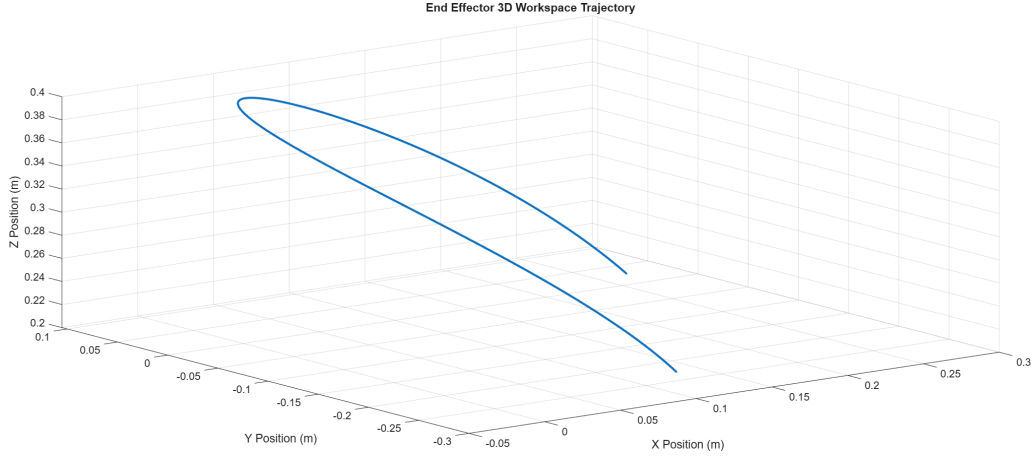


Figure 3.2: Obtained 3D Workspace Trajectory from Harmonic Inputs

3.2 Dynamic Environmental Collision

Upon contacting the environmental table at $Z = 0.30$ m, the unactuated lower arm buckled organically, with the contact force block successfully registering the impact normal force spike.

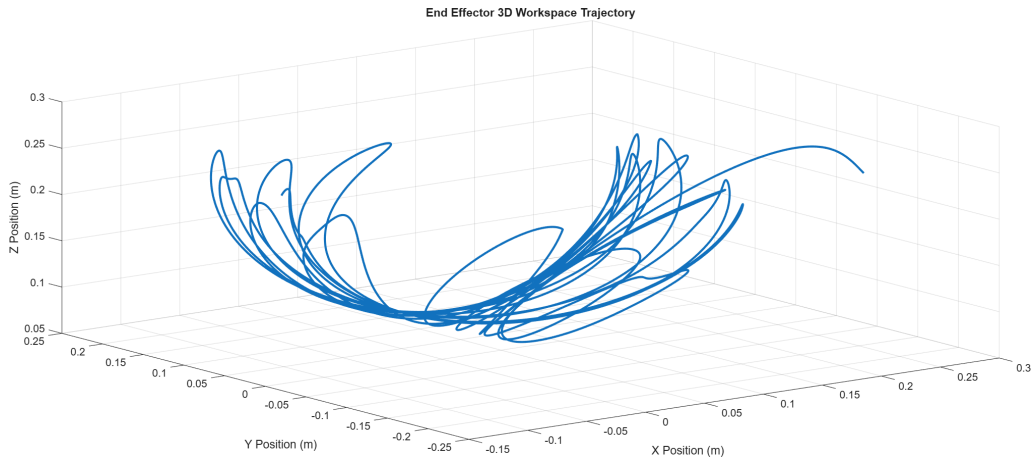


Figure 3.3: Obtained Trajectory Exhibiting Physical Buckling During Table Impact

3.3 Smooth Trajectory Generation and Motion Tracking

The second-order filtering eliminated position violations, converting raw steps into S-curves. The accompanying scope and motion tracking outputs verify smooth acceleration and deceleration without torque singularities.

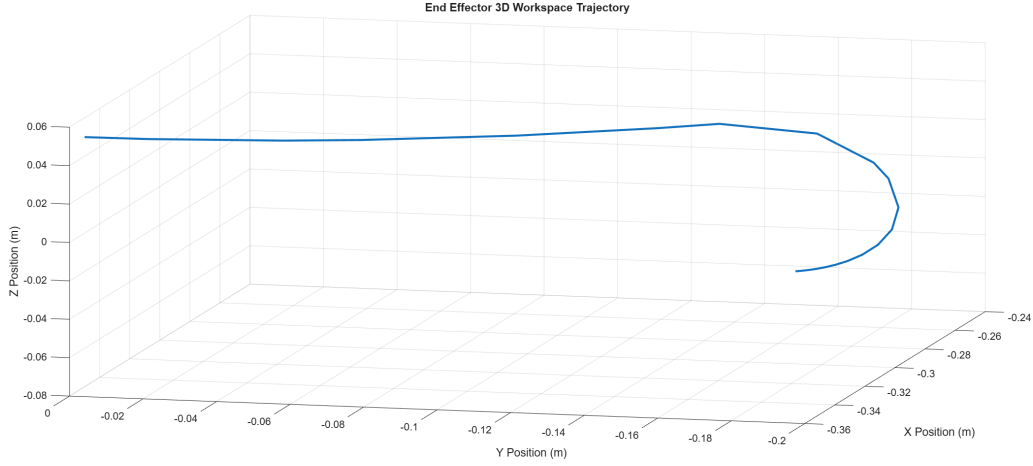


Figure 3.4: Obtained S-Curve Spatial Trajectory and Motion Tracking Scope Output from Filtered Step Inputs

3.4 Kinematic Validation (D-H Parameters)

The physical lengths derived from the solid blocks and joint offsets match standard Denavit-Hartenberg conventions.

Table 3.1: D-H Parameters of the 4-DOF Manipulator

Link (i)	a_i (Length)	α_i (Twist)	d_i (Offset)	θ_i (Angle)
1 (Base)	0	90°	d_1	θ_1
2 (Shoulder)	a_2	0°	0	θ_2
3 (Elbow)	a_3	0°	0	θ_3
4 (Wrist)	0	0°	0	θ_4

Explicit Parameter Values :

- **Link Offsets (d_i):** $d_1 = 0.10$ m (Base length, L_{base}); $d_2 = 0$ m, $d_3 = 0$ m, $d_4 = 0$ m.
- **Link Lengths (a_i):** $a_2 = 0.20$ m (Upper arm length, L_{upper}); $a_3 = 0.10$ m (Lower arm length, L_{lower}); $a_1 = 0$ m, $a_4 = 0$ m.

Chapter 4

CONCLUSION AND FUTURE SCOPE

The customized 4-DOF manipulator model generated in Simscape Multibody successfully passed all foundational kinematic and dynamic tests required for aerospace prototyping. The sine wave inputs validated the functional workspace, convex hull mechanics handled collision interactions, and second-order filtering prevented infinite torque singularities.

Future Scope: Moving forward in the Def-Space development cycle, the static U-shaped end effector should be split into two separate geometric bodies connected by a prismatic joint to execute complete automated payload Pick-and-Place operations. Furthermore, integrating a PID controller block will allow for the simulation of realistic motor payload strain.

Chapter 5

APPENDICES

Appendix A: MATLAB Plotting Script

```
% Script to plot out.trajectory
x_data = out.trajectory.signals.values(:,1);
y_data = out.trajectory.signals.values(:,2);
z_data = out.trajectory.signals.values(:,3);

figure;
plot3(x_data, y_data, z_data, 'b-', 'LineWidth', 2);
grid on;
xlabel('X Position (m)');
ylabel('Y Position (m)');
zlabel('Z Position (m)');
title('End Effector 3D Workspace Trajectory');
```